A company needs help updating their file system. You must create a simple file-manipulation program that performs the following actions:

- Checks to see whether a file exists.
- If the file exists, displays its contents.

Complete the code by selecting the correct code segment from each drop-down list.

Note: You will receive partial credit for each correct selection.

Answer Area

```
import os
if [os.path.isfile('myFile.txt'):]
    file = open('myFile.txt')
    [print(file.read())]
    file.close()
```

A bicycle company is creating a program that allows customers to log the number of miles biked. The program will send messages based on how many miles the customer logs.

You write the following Python code. Line numbers are included for reference only.

```
01
02      name = input("What is your name? ")
03      return name
04
05      calories = miles * calories_per_mile
06      return calories
07 distance = int(input("How many miles did you bike this week? "))
08 burn_rate = 50
09 biker = get_name()
10 calories_burned = calc_calories(distance, burn_rate)
11 print(biker, ", you burned about ", calories_burned, " calories.")
```

You need to define the two required functions.

Which two code segments should you use for line 01 and line 04? Each correct answer presents part of the solution. (Choose 2.)

- ☒ A. `01 def get_name():`
- ☐ B. `01 def get_name(biker):`
- ☐ C. `01 def get_name(name):`
- ☐ D. `04 def calc_calories():`
- ☒ E. `04 def calc_calories(miles, burn_rate):`
- ☐ F. `04 def calc_calories(miles, calories_per_mile):`

Review the following code segment:

```
f = open("python.txt", "a")
f.write("This is a line of text.")
f.close()
```

For each statement about the code segment, select **True** or **False**.

Answer Area

	True	False
A file named `python.txt` is created if it does not exist.	◉	○
The data in the file will be overwritten.	○	◉
Other code can open the file after this code runs.	◉	○

Question 33 of 35

You write the following function to read a data file and print each line of the file. Line numbers are included for reference only.

```
01 def read_file(file):
02     line = None
03     if os.path.isfile(file):
04       data = open(file,'r')
05       for line in data:
06         print(line)
```

When you run the program, you receive an error on line 03.

What is causing the error?

- A. The `isfile` method does not exist in the `path` object.
- B. The `isfile` method does not accept one parameter.
- C. The `path` method does not exist in the `os` object.
- D. You need to import the `os` library.

Question 30 of 35

Review the following code segment:

```
product = 2
n = 5
while (n != 0):
    product *= n
    print (product)
    n -= 1
    if n == 3:
       break
```

How many lines of output does the code print?

Enter the number as an integer here: 2

You are writing code to meet the following requirements:

- Allow users to repeatedly enter words.
- Output the number of characters in each word.

Complete the code by selecting the correct option from each drop-down list.

Note: You will receive partial credit for each correct selection.

Answer Area

```
x = "Hello"
[while] x != "QUIT":
    num = 0
    [for] char [in] x:
        num += 1
    print(num)
    x = input("Enter a new word or QUIT to exit: ")
```

You are creating a program that accepts user input. The program must cast the input into an integer, and properly handle the error if it cannot do so.

Complete the code by selecting the correct code segment from each drop-down list.

Answer Area

```
while True:
    try:
        x = int(input("Please enter a number: "))
        break
    except ValueError:
        print("Not a valid number.  Try again...")
```

You are writing a Python program to determine if a number (`num`) the user inputs is one, two, or more than two digits (`digits`).

Complete the code by selecting the correct code segment from each drop-down list.

Answer Area

```
num = int(input("Enter a number with 1 or 2 digits: "))
digits = "0"
[if num > -10 and num < 10:]
    digits = "1"
[elif num > -100 and num < 100:]
    digits = "2"
[else:]
    digits = ">2"
print(digits + " digits.")
```

You work on a team that is developing a game.

You need to write code that generates a random number that meets the following requirements:

- The number is a multiple of 5.
- The lowest number is 5.
- The highest number is 100.

Which two code segments will meet the requirements? Each correct answer presents a complete solution. (Choose 2.)

Note: You will receive partial credit for each correct answer.

☐ A.
```
from random import randrange
print(randrange(0, 100, 5))
```

☐ B.
```
from random import randint
print(randint(0, 20) * 5)
```

☒ C.
```
from random import randrange
print(randrange(5, 105, 5))
```

☒ D.
```
from random import randint
print(randint(1, 20) * 5)
```

You are creating a Python program that compares numbers.

You write the following code. Line numbers are included for reference only.

```
01 num1 = eval(input ("Please enter the first number: "))
02 num2 = eval(input ("Please enter the second number: "))
03 if num1 == num2:
04     print("The two numbers are equal.")
05 if num1 <= num2:
06     print("Number 1 is less than number 2.")
07 if num1 > num2:
08     print("Number 1 is greater than number 2.")
09 if num2 = num1:
10     print("The two numbers are the same.")
```

You need to ensure that the comparisons are accurate.

For each statement, select **True** or **False**.

Note: You will receive partial credit for each correct selection.

Answer Area

	True	False
The `print` statement at line 04 will print only if the two numbers are equal in value.	◉	○
The `print` statement at line 06 will print only if `num1` is less than `num2`.	○	◉
The `print` statement at line 08 will print only if `num1` is greater than `num2`.	◉	○
The statement at line 09 is an invalid comparison.	○	◉

False

For each statement about `try` statements, select **True** or **False**.

Answer Area

	True	False
A `try` statement can have one or more `except` clauses.	◉	○
A `try` statement can have a `finally` clause without an `except` clause.	◉	○
A `try` statement can have a `finally` clause and an `except` clause.	◉	○
A `try` statement can have one or more `finally` clauses.	○	◉

You are writing a function that increments the player score in a game.

The function has the following requirements:

- If no value is specified for `points`, then `points` start at one.
- If bonus is `True`, then `points` must be doubled.

You write the following code. Line numbers are included for reference only.

```
01 def increment_score(score, bonus, points):
02     if bonus == True:
03         points = points * 2
04     score = score + points
05     return score
06 points = 5
07 score = 10
08 new_score = increment_score(score, True, points)
```

For each statement, select **True** or **False**.

Note: You will receive partial credit for each correct selection.

Answer Area

	True	False
To meet the requirements, you must change line 01 to: `def increment_score(score, bonus, points = 1):`	◉	○
If you do not change line 01 and the function is called with only two parameters, an error occurs.	◉	○
Line 03 will also modify the value of the variable `points` declared at line 06.	○	◉

You are creating a function to calculate admission fees (`admission_fee`) based on the following rules:

- Anyone under age 5 = free admission
- Anyone age 5 or older who is in school = $10
- Anyone age 5 to 17 who is not in school = $20
- Anyone older than age 17 who is not in school = $50

`rate` represents the price in dollars.

Complete the code by selecting the correct code segment from each drop-down list.

Answer Area

```
def admission_fee(age, school):
    rate = 0
    [if age >= 5 and school == True:]
        rate = 10
    [elif age >= 5 and school == False:]
        [if age <=17:]
            rate = 20
        else:
            rate = 50
    return rate
```

Question 24 of 35

You develop a Python application for your company.

You want to add notes to your code so other team members will understand it.

What should you do?

A. Place the notes within <!-- and --> in any code segment.

B. Place the notes within /* and */ in any code segment.

C. Place the notes after # on any line.

D. Place the notes after // on any line.

You are creating a Python program that will let a user guess a number from 1 to 10. The user is allowed up to three guesses. If the user guesses the correct number, the program must stop asking for guesses.

Complete the code by moving the appropriate code segments from the list on the left to the correct locations on the right. You may use each code segment once, more than once, or not at all.

Code Segments

`break`	`chance += 1`	`chance = 2`
`pass`	`while chance < 3`	`while chance < 3:`
`while chance <= 3:`		

Answer Area

```
from random import randint
target = randint(1,10)
chance = 1
print ("Guess an integer from 1 to 10. You will have 3 chances.")
```

`while chance <= 3:`

```
    guess = int(input("Guess an integer: "))
    if guess > target:
      print ("Guess is too high")
    elif guess < target:
      print ("Guess is too low")
    else:
      print ("Guess is just right!")
```

`break`

`chance += 1`

Target area for the third prompt.

You are writing a Python application that includes multiple operations on the same line of code. You need to determine the correct order of operations.

Move the type of operation from the list on the left to the correct locations on the right, with the type of operation that will be performed first at the top and the type of operation that will be performed last at the bottom.

Note: You will receive partial credit for each correct response.

Operation Types

- Addition and Subtraction
- And
- Exponents
- Multiplication and Division
- Parentheses
- Unary positive, negative, not

Operation Types in Order

Operation type performed first

- Parentheses
- Exponents
- Unary positive, negative, not
- Multiplication and Division
- Addition and Subtraction

Operation type performed last

- And

Target area for the operation type performed last.

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You need to identify the data types of various type operations.

Move the appropriate data types from the list on the left to the correct type operations on the right. You may use each data type once, more than once, or not at all.

Note: You will receive partial credit for each correct match.

Data Types

int | float | str | bool

Answer Area

type(+1E10)	float
type(5.0)	float
type("True")	str
type(False)	bool

You need to test whether an object is an instance of a specific class.

How should you set up the unit test?

Complete the code by selecting the correct option from each drop-down list.

Note: You will receive partial credit for each correct selection.

Answer Area

```
import [unittest]
class TestIsInstance( [unittest.TestCase] ):
    def [test_isInstance(self):]
        [self.assertIsInstance(obj, cls, msg=None)]
if __name__ == '__main__':
    unittest.main()
```

You are building a Python program that displays all prime numbers from 2 to 100. (A prime number is any number that is divisible only by itself and 1.)

Complete the code by moving the appropriate code segments from the list on the left to the correct locations on the right.

Note: You will receive partial credit for each correct code placement.

Code Segments

```
break
```

```
continue
```

```
p = p + 1
```

```
p = 2
is_prime = True
while p <= 100:
```

```
p = 2
while p <= 100:
    is_prime = True
```

Answer Area

```
p = 2
while p <= 100:
    is_prime = True
    for i in range(2, p):
        if p % i == 0:
            is_prime = False
            break
if is_prime == True:
    print(p)
p = p + 1
```

Target area for the fourth code segment.

Review the following code:

```
x = "oranges"
y = "apples"
z = "bananas"
data = "{1} and {0} and {2}"
print(data.format(z, y, x))
```

What is the output of the `print` statement?

- A. `apples and oranges and bananas`
- B. `apples and bananas and oranges`
- C. `oranges and apples and bananas`
- D. `bananas and oranges and apples`

You are creating an eCommerce script that accepts input from the user and outputs the data in a comma-delimited format.

You write the following code to accept input:

```
item = input("Enter the item name: ")
sales = int(input("Enter the quantity: "))
```

The output must meet the following requirements:

- Enclose strings in double quotes.
- Do not enclose numbers in quotes or other characters.
- Separate items by commas.

You need to complete the code to meet the requirements.

Which two code segments could you use? Each correct answer presents a complete solution. (Choose 2.)

Note: You will receive partial credit for each correct answer.

☒ A. `print('"{0}",{1}'.format(item, sales))`

☐ B. `print(item + ',' + sales)`

☐ C. `print("{0},{1}".format(item, sales))`

☒ D. `print('"' + item + '",' , sales)`

You are writing a program to randomly assign rooms (`room_number`) and team-building groups (`group`) for a company retreat.

Complete the code by selecting the correct code segment from each drop-down list.

Note: You will receive partial credit for each correct selection.

Answer Area

```
import random
roomsAssigned=[1]
room_number=1
groupList=["Ropes","Rafting","Obstacle","Wellness"]
count=0
print("Welcome to CompanyPro's Team-Building Weekend!")
name=input("Please enter your name (q to quit)? ")
while name.lower() != 'q' and count < 50:
    while room_number in roomsAssigned:
        [room_number=random.randint(1,50)]
    print(f"{name}, your room number is {room_number}")
    roomsAssigned.append(room_number)
    [group = random.choice(groupList)]
    print(f"You will meet with the {group} Group this afternoon.")
    count+=1
    name=input("Please enter your name (q to quit)? ")
```

You develop a Python application for your company.

You need to complete the code so that the `print` statements are accurate.

Complete the code by selecting the correct code segment from each drop-down list.

Answer Area

```
numList = [1, 2, 3, 4, 5]
alphaList = ["a", "b", "c", "d", "e"]
if numList == alphaList :
    print("The values in numList are equal to alphaList")
else :
    print("The values in numList are not equal to alphaList")
```

You create the following Python function to calculate the power of a number. Line numbers are included for reference only.

```
01 # The calc_power function calculates exponents
02 # x is the base
03 # y is the exponent
04 # The value of x raised to the y power is returned
05 def calc_power(x, y):
06   comment = "# Return the value"
07   return x ** y # raise x to the y power
```

For each statement, select **True** or **False**.

Note: You will receive partial credit for each correct selection.

Answer Area

	True	False
Python will not check the syntax of lines 01 through 04.	◉	○
The pound sign (#) is optional for lines 02 and 03.	○	◉
The string in line 06 will be interpreted as a comment.	○	◉
Line 07 contains an inline comment.	◉	○



The `Script.py` file contains the following code:

```
import sys
print(sys.argv[2])
```

You run the following command:

```
python Script.py Cheese Bacon Bread
```

What is the output of the command?

- A. `Script.py`
- B. `Bread`
- C. `Bacon`
- D. `Cheese`

A coworker wrote a program that inputs names into a database. Unfortunately, the program reversed the letters in each name.

You need to write a Python function that outputs the characters in a name in the correct order.

Complete the code by selecting the code segment from each drop-down list.

Note: You will receive partial credit for each correct selection.

Answer Area

```
#Function reverses characters in a string.
#returns new string in reversed order.

def reverse_name(backward_name):
    forward_name = ""
    length = [len(backward_name)-1]
    while length >= 0:
        forward_name += [backward_name[length]]
        length = length-1
    return forward_name

print(reverse_name("nohtyp"))
```

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For each statement about the following function, select **True** or **False**.

```
def
grosspay(hours=40,rate=25,pieces=0,piecerate=0,salary=0):
    overtime=0
    if pieces > 0:
        return pieces * piecerate
    if salary > 0:
        pass
    if hours > 40:
        overtime = (hours-40) * (1.5 * rate)
        return overtime + (40 * rate)
    else:
        return hours * rate
```

Answer Area

	True	False
A function call of `grosspay()` will create a syntax error.	○	◉
A function call of `grosspay(salary=50000)` will return nothing.	◉	○
A function call of `grosspay(pieces=500, piecerate=4)` will return a result of 2000.	◉	○

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Question 9 of 35

You write the following code to determine a student's final grade based on their current grade (`grade`) and rank (`rank`):

```
grade = 76
rank = 3

if grade > 80 and rank >= 3:
   grade += 10
elif grade >= 70 and rank > 3:
   grade += 5
else:
   grade -= 5

print(grade)
```

What value will print?

- A. 71
- B. 76
- C. 81
- D. 86

A game development company needs a way to find the number of words in a list that contain a specific letter.

Complete the code by selecting the correct code segment from each drop-down list.

Note: You will receive partial credit for each correct selection.

Answer Area

```
# Function accepts list of words and letter to search for.
# Returns count of the number of words that contain that letter.
def count_letter(letter, word_list):
    count = 0

    for [word in word_list:]
        if [letter in word:]
            count += 1
    return count

# word_list is populated by the readWords() function. Code not shown.
word_list = readWords()

letter = input("Which letter would you like to count")
letter_count = count_letter(letter, word_list)
print("There are: ", letter_count, " words that contain ", letter)
```

Question 5 of 35

What does the following statement do?

```
data = input()
```

- A. Creates an HTML input element
- B. Allows a user to enter text in the console
- C. Displays a message box that allows user input
- D. Displays all input peripheral devices on the computer

Question 6 of 35

A company needs help updating their file system. You must create a simple file-manipulation program that performs the following actions:

- Creates a file using the specified name.
- Appends the phrase "`End of listing`" to the file.

You need to complete the code to meet the requirements.

Complete the code by selecting the correct code segment from each drop-down list.

Note: You will receive partial credit for each correct selection.

Answer Area

```
import os
file = open('myFile.txt', 'a')
    file.write ("End of listing")
file.close()
```

You need to identify the results of performing various slicing operations on the following sequence structure:

```
alph = "abcdefghijklmnopqrstuvwxyz"
```

Move the appropriate results from the list on the left to the correct slicing operations on the right. You may use each result once, more than once, or not at all.

Note: You will receive partial credit for each correct match.

Results

def	abcdef	cde
defg	cdef	abcde

Answer Area

alph[3:6]	def
alph[:6]	abcdef

Target area for the operation alph[:6]

A friend asks you to refactor and document the following Python code:

```
value1 = 9
value2 = 4

answer = (value1 % value2 * 10) // 2.0 ** 3.0 + value2
```

You run the code.

What is the result?

A. The value 5.667 is displayed.

B. The value 5.0 is displayed.

C. A syntax error occurs.

D. The value 129 is displayed.

The following function calculates the value of an expression that uses an exponent. Line numbers are included for reference only.

```
01 def calc_power(a, b):
02   return a**b
03 base = input("Enter the number for the base: ")
04 exponent = input("Enter the number for the exponent: ")
05 result = calc_power(base, exponent)
06 print("The result is " + result)
```

For each statement, select **True** or **False**.

Note: You will receive partial credit for each correct selection.

Answer Area

	True	False
The code will generate an error in line 03 and line 04.	○	◉
The code will generate an error in line 02 and line 05.	◉	○
The code will correctly output data to the console.	○	◉

You find errors while evaluating the following code. Line numbers are included for reference only.

```
01 numbers = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
02 index = 0
03 while (index < 10)
04     print(numbers[index])
05
06     if numbers(index) = 6
07         break
08     else :
09         index += 1
```

You need to correct the code at line 03 and line 06.

Evaluate the code and answer the questions by selecting the correct option from each drop

Answer Area

Which code segment should you use at line 03? while (index < 10) :

Which code segment should you use at line 06? if numbers[index] == 6 :

Answer options for the second question.